



PRE-BOOTCAMP READINESS SESSION

# AI in Biomed Boot Camp

Getting everyone set up on AWS — GW Research Gateway

**60 minutes** · Rocky Linux 8 Remote Desktop · Jupyter + Ollama

# Why we're on AWS today

## The situation

- Pegasus GPFS storage failed and won't be fixed before our fixed dates.
- Rescheduling isn't an option — our job is to make it work now.
- GW Research Gateway (AWS) gives everyone a full Linux desktop today.
- Pegasus was always a vehicle, not the point — the AI/biomed skills are.

## What carries over

- ✓ Jupyter notebooks
- ✓ PyTorch / Torch
- ✓ Ollama + local models
- ✓ bash / CLI workflows
- ✓ Python environments (venv, pip)

*Only Slurm / HPC-scheduler specifics don't transfer.*

## RUN OF SHOW

# The hour at a glance

*Slow steps (approval, ~15-min launch, model download) run in the background while we teach.*

0:00–0:05

### Welcome + launch instances

Kick off the ~15-min provisioning clock immediately

0:05–0:20

### While it spins up — teach

Cloud at GW · access model · Linux basics · Google Colab vs Jupyter

0:20–0:35

### Hands-on setup

Log into the desktop, stand up Jupyter, start the model pull

0:35–0:50

### Notebooks + Ollama

Run ollama\_aws.ipynb · models · when to use notebooks

0:50–1:00

### Wrap-up + readiness check

Troubleshooting · RTShelp / Office Hours · confirm everyone's green

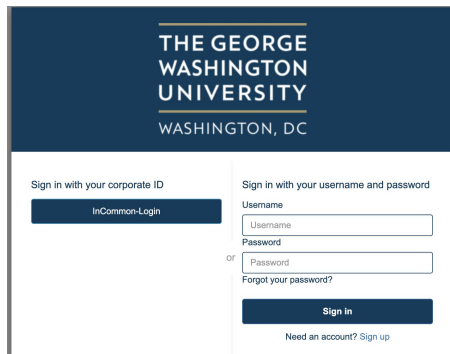
# Launch first, talk second

0:00–0:05

Screen-share this click-path — everyone launches together, then we teach while it builds.

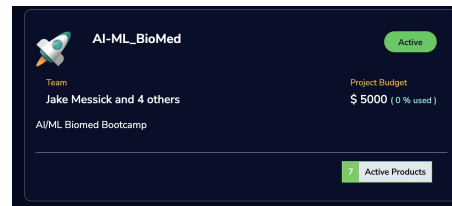
## 1 Log in

rg.arc.gwu.edu → InCommon → Sign in with GW



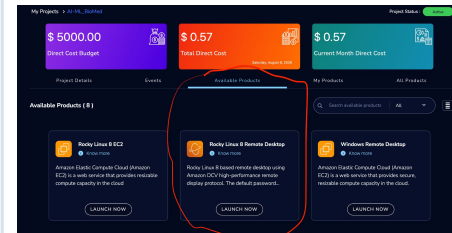
## 2 Pick the project

Select AI-ML\_BioMed



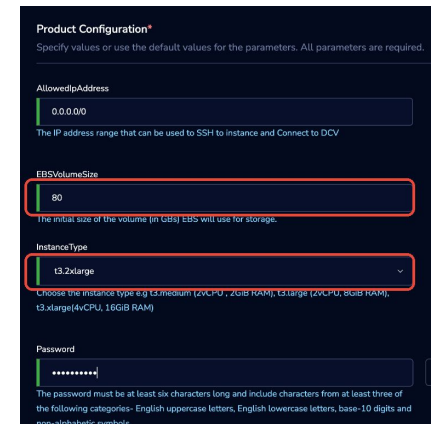
## 3 Pick the product

Rocky Linux 8 Remote Desktop



## 4 Configure + launch

80 GB · t3.2xlarge · Launch Now



≈15 minutes to provision. We start every attendee's instance now — no one sits watching a spinner.

THE ONE THING THAT CAN'T WAIT

# Admin approval is the hidden blocker

**First-ever login shows "requires admin approval."** That's normal — but until it's granted, an attendee can't launch anything. Handle it before the clock matters.

1

**Have attendees do their FIRST login the day before**

So approvals are already granted when the session starts.

2

**Approver on standby at minute 0**

Someone with admin rights ready to approve any stragglers live.

3

**Confirm everyone is on AI-ML\_BioMed beforehand**

No project membership = nothing to launch.

# While it spins up ( $\approx 15$ min of teaching)

0:05–0:20

## 1 Where cloud fits at GW

Pegasus (HPC/GPU), Raptor (HTC/HTCondor), Research Cloud (persistent). Why today's skills transfer to all of them.

## 2 How access works

InCommon sign-in, projects & budgets, and the \$5,000 AI-ML\_BioMed project you just launched into.

## 3 Linux & the terminal

What a shell/bash is, tab completion, command history — and why CLI still runs HPC and cloud.

## 4 Notebooks & Google Colab

What notebooks are good for, and how Jupyter compares to Colab (next slide).

# Google Colab vs. Jupyter on your cloud desktop

## Google Colab

- Zero setup — runs in the browser
- Free (limited) GPU, Google-hosted
- Great for quick demos and sharing
- Sessions time out; data lives in Drive
- Limited control over the environment

## Jupyter on the cloud desktop

- Full control of packages & environment
- Persistent project storage
- Your own compute (t3.2xlarge)
- Same workflow you'll use in the bootcamp
- Mirrors real HPC / research habits

*Bottom line: Colab is perfect for trying an idea; the cloud desktop mirrors what you'll actually run on.*

# Get into the desktop and stand up Jupyter

0:20–0:35

## Log into the desktop

- A new browser window opens with your desktop.
- Log in with the password you created at launch.
- Tip: click in the window and drag upward to reveal the password box.

## Then, in a terminal

```
sudo dnf install python3.12 -y
python3.12 -m venv my_new_env
source my_new_env/bin/activate
pip install notebook
jupyter notebook
```

## Start the model pull early

As soon as Jupyter opens, open `ollama_aws.ipynb` and run the model-pull cells right away.

The download runs in the background while we talk through notebooks — so the model is ready before the bootcamp, not during it.

### Full path:

```
/home/ec2-user/studies/ProjectStorage/ollama_aws.ipynb
```



# Notebooks + Ollama

0:35–0:50

## Notebooks

- Run ollama\_aws.ipynb end to end.
- What notebooks are good for — and when a plain script is better.
- How notebooks relate to normal shell workflows.
- Same interface you'll use throughout the bootcamp.

## Ollama & models

- Calling Ollama from inside a notebook.
- Available models: frontier vs. open-weight.
- Size & quantization trade-offs (speed vs. quality vs. memory).
- The background pull finishes here — confirm the model responds.

# When things go wrong — and where to get help

0:50–1:00

## Login fails

Re-check InCommon / GW sign-in; first login needs admin approval.

## Instance won't launch

Confirm AI-ML\_BioMed membership, budget, and 80 GB / t3.2xlarge config.

## Jupyter won't start

Activate the venv first (source my\_new\_env/bin/activate), then re-run.

## Package or model missing

Re-run the pip / model-pull cells; check the venv is active.

## Where to get help

RTShelp · Office Hours — bring access working, don't wait until the bootcamp.

## Watch the budget

t3.2xlarge bills against the \$5,000 project budget. STOP your instance when you're done for the day.

## READINESS CHECK

# You're ready when...



Logged into Research Gateway (rg.arc.gwu.edu)



Launched Rocky Linux 8 Remote Desktop (80 GB · t3.2xlarge)



Jupyter running from your `venv`

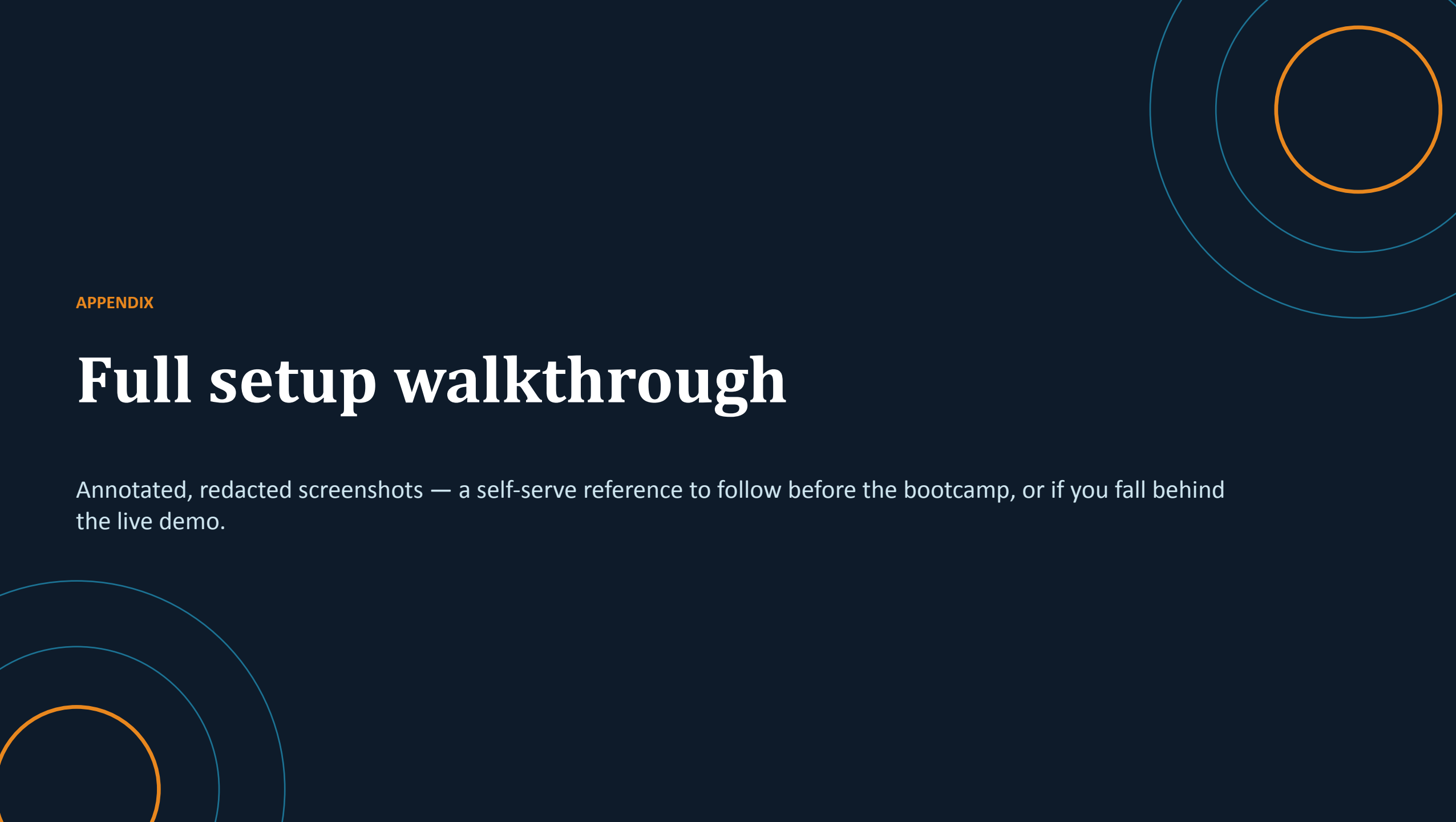


Ran `ollama_aws.ipynb` end to end



Model pulled and confirmed responding in a notebook

*Come to the bootcamp with all five green.*

The slide features a dark blue background with decorative elements consisting of concentric circles in the corners. In the top right, there are three concentric circles: the innermost is orange, the middle is a lighter blue, and the outermost is a very light blue. In the bottom left, there are also three concentric circles: the innermost is orange, the middle is a lighter blue, and the outermost is a very light blue.

## APPENDIX

# Full setup walkthrough

Annotated, redacted screenshots — a self-serve reference to follow before the bootcamp, or if you fall behind the live demo.

# Sign in to Research Gateway

## 1 Go to [rg.arc.gwu.edu](https://rg.arc.gwu.edu)

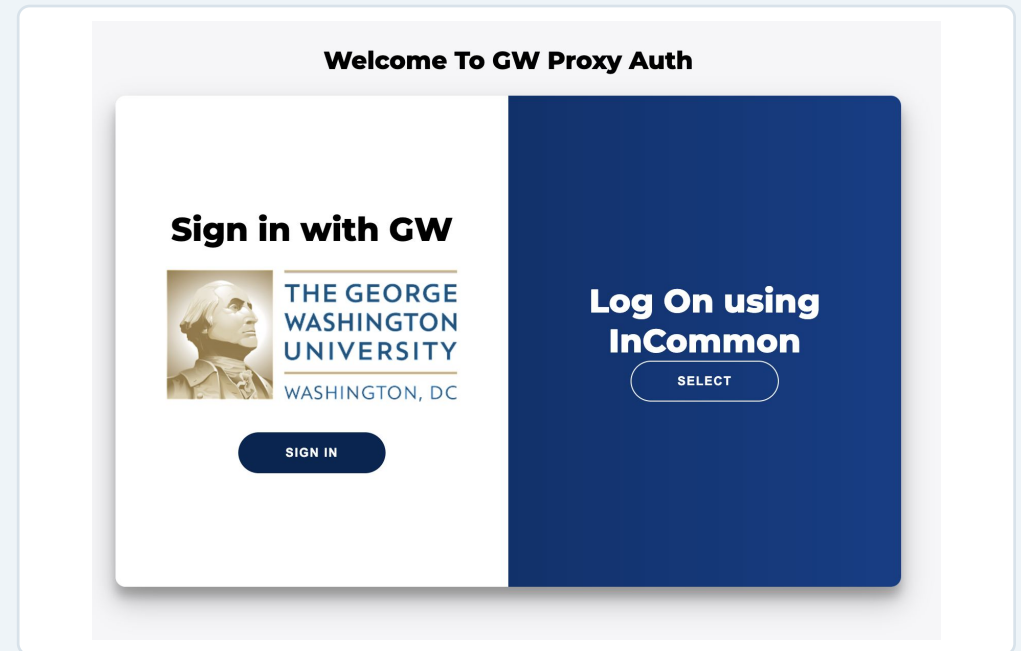
Choose InCommon-Login.



The screenshot shows the login page for The George Washington University Research Gateway. The header features the university's name and location. Below the header, there are two main login options: 'Sign in with your corporate ID' and 'Sign in with your username and password'. The 'Sign in with your corporate ID' option has a button labeled 'InCommon-Login'. The 'Sign in with your username and password' option includes fields for 'Username' and 'Password', a 'Sign in' button, and a link for 'Forgot your password?'. At the bottom, there is a link for 'Need an account? Sign up'.

## 2 Sign in with GW

First-ever login shows “requires admin approval” — that's normal; do this the day before.

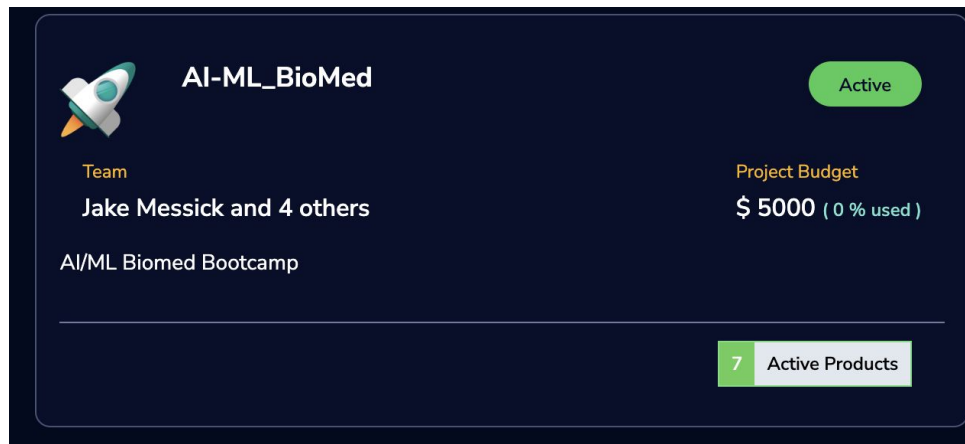


The screenshot shows the 'Welcome To GW Proxy Auth' page. It features a central card with the title 'Sign in with GW' and the university's logo. Below the logo is a 'SIGN IN' button. To the right of the card, there is a dark blue sidebar with the text 'Log On using InCommon' and a 'SELECT' button. The overall layout is clean and professional, with a focus on the login process.

# Open the project, pick the product

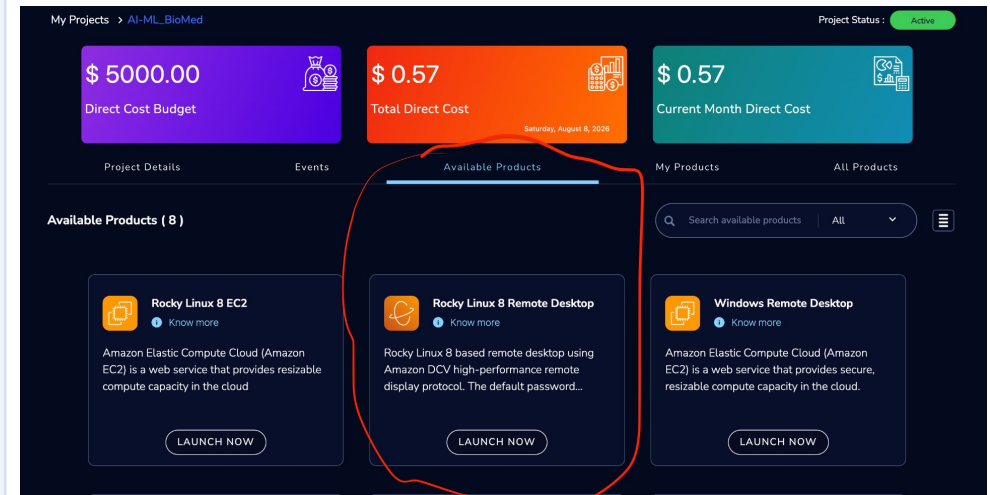
## 3 Open AI-ML\_BioMed

From My Projects, open the AI-ML\_BioMed project card.



## 4 Available Products → Rocky Linux 8 Remote Desktop

Under Available Products, choose Rocky Linux 8 Remote Desktop (circled).



# Configure, launch, and log in

## 5 Set storage, instance type, password

EBSVolumeSize 80 · InstanceType t3.2xlarge · choose a password → Launch Now.

**Product Configuration\***  
Specify values or use the default values for the parameters. All parameters are required.

**AllowedIpAddress**  
0.0.0.0/0  
The IP address range that can be used to SSH to instance and Connect to DCV

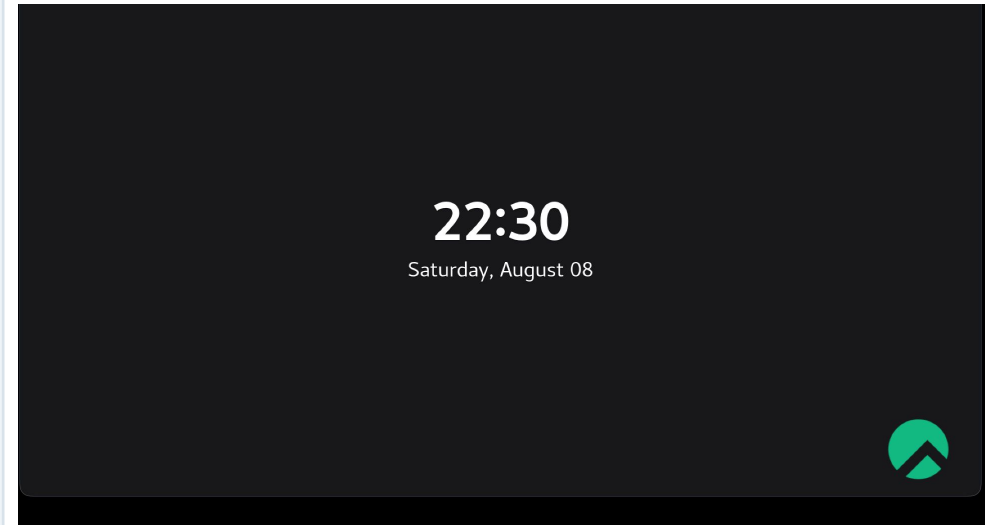
**EBSVolumeSize**  
80  
The initial size of the volume (in GBs) EBS will use for storage.

**InstanceType**  
t3.2xlarge  
Choose the instance type e.g t3.medium (2vCPU, 2GiB RAM), t3.large (2vCPU, 8GiB RAM), t3.xlarge(4vCPU, 16GiB RAM)

**Password**  
.....  
The password must be at least six characters long and include characters from at least three of the following categories- English uppercase letters, English lowercase letters, base-10 digits and non-alphabetic symbols.

## 6 Log into the desktop

Opens in a new browser tab. Log in with your password (click and drag upward to reveal the box).



# Stand up Jupyter and pull the model

Open a terminal on the desktop and run:

```
sudo dnf install python3.12 -y
python3.12 -m venv my_new_env
source my_new_env/bin/activate
pip install notebook
jupyter notebook
```

Then open and run:

```
/home/ec2-user/studies/ProjectStorage/ollama_aws.ipynb
```

Run the model-pull cells first so the download finishes in the background.

*A Jupyter notebook opens in a browser on the desktop. The venv must be active before you run jupyter notebook.*